Autonomous Next-Generation Aerial Refueling System (ANGARS)

Functional Analysis Report

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Course: 645.800 - Systems Engineering Masters Capstone

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1 Report Introduction

1.1 Scope

This report intends to;

- Provide updates to ANGARS Structural Design by highlighting key changes between proposal structural definition and changes that were made during functional analysis and development.
- Provide a baseline definition of Cameo modeling profile by highlighting common and custom profiles that will support Capstone expectations.

- Provide a baseline of system functional analysis by detailing top level and decomposed behaviors of the system and their interactions with stakeholders.

1.2 ANGARS Structural Design Updates

1.2.1 Context Diagram Updates

Pictured below is an updated Whitebox Context Diagram. As system design matures information on this diagram will also mature to show changes in stakeholder interactions with the system and how the system is designed to handle some of these changes. A whitebox was provided over a black box to better show the traceability of changes made between ANGARS and interfacing actors, and how that flows into the systems internals. The whitebox design has stayed relatively fixed in terms of general subsystem definition, as this will be decomposed further during the Conceptual Design Report.

Key changes from last update (proposal) include;

- Matured signal definition by adding preliminary attributes to signals to capture scope of I/O throughout the system context
- Control interface definition between ANGARS and Control Center now has an available Relay Satellite component. This is to better align with system requirements and OV-1 to add to system adversarial resiliency capabilities by providing an avenue of communications to the ground in the event of direct line of site or EW jamming disturbances.
- Better definition of interactions between receiving aircraft and ANGARS. Updates include definition to key messages such as Tanker Positioning Support and Scheduling message information, which can be found in the table below. This design maturation now better defines what will be communicated between receiving aircraft to support a refueling mission.
- Removal of "Support Systems" actor, the actor through which software and firmware updates flowed into ANGARS from. This data flow has been reallocated to flow between the Control Center and ANGARS, as this better aligns with mission capabilities. Ground control/central commanding will be responsible for software/firmware uploads to ensure proper security channels constraints can be applied to a single over the air (OTA) interface, reducing complexity and ensuring resilience.
- Updates to Maintainer interfacing with the system. This includes reallocation of software/firmware updates to this physical interfacing with the system, to support the same needs as the bullet above.

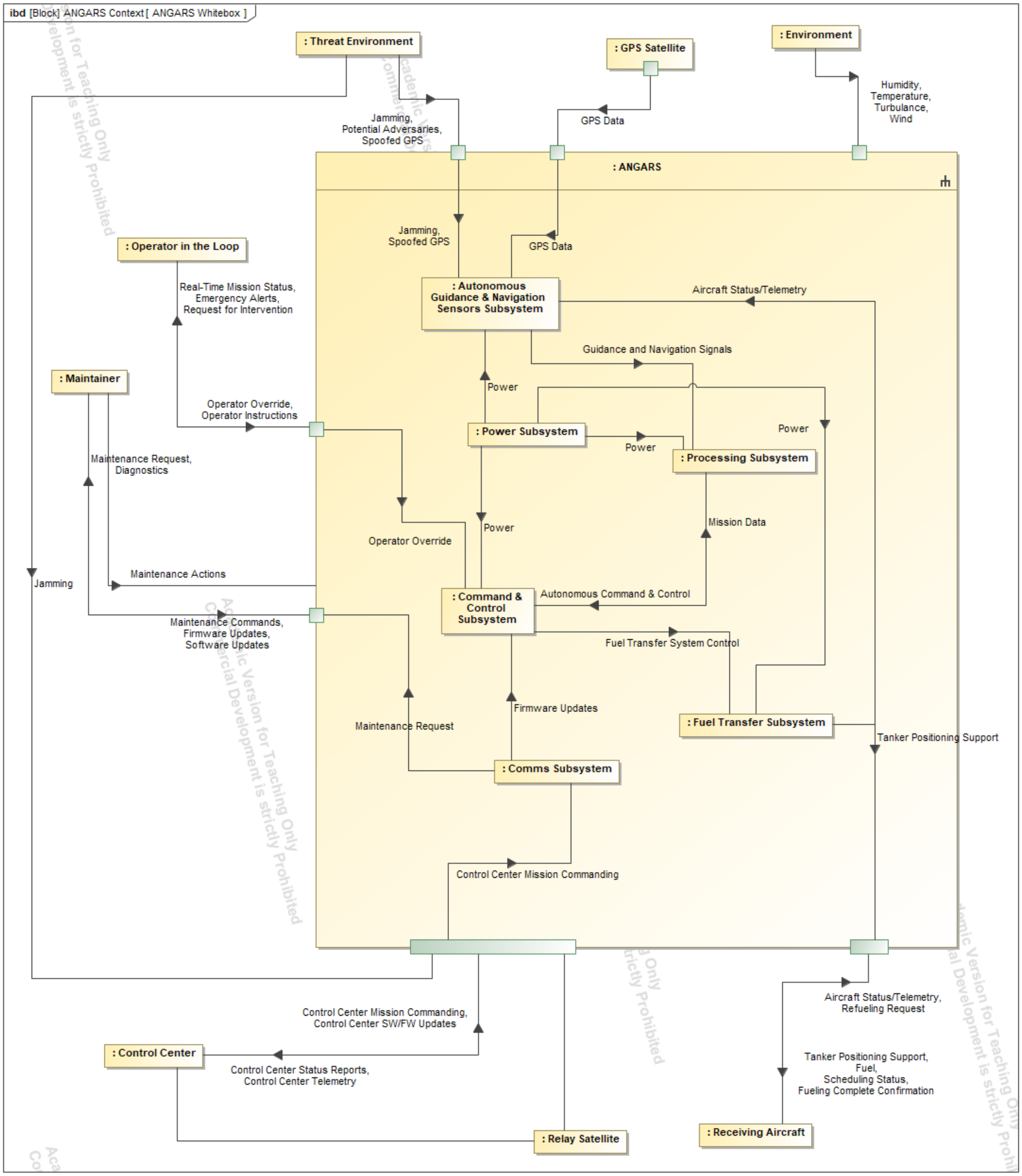


Figure 1: ANGARS FAR Whitebox

1.2.2 Item Flow Definition

#	Name	Owned Attribute
1	Aircraft Status/Telemetry	+Fuel Level +Position +Speed
2	Control Center Mission Commanding	
3	Control Center Status Reports	
4	Control Center SW/FW Updates	
5	Control Center Telemetry	+Mission Progress +Logs +Situation Updates
6	Diagnostics	+Diagnostics Data +Fault Logs +Health & Status Reports
7	Emergency Alerts	
8	Firmware Updates	
9	Fuel	
10	Fueling Complete Confirmation	
11	GPS Data	
12	Jamming	
13	Maintenance Actions	
14	Maintenance Commands	
15	Maintenance Request	
16	Operator Instructions	
17	Operator Override	
18	Potential Adversaries	
19	Real-Time Mission Status	+Mission Status +Daignostics +Alerts
20	Refueling Request	+ID +Fuel Type +Priority Indicator
21	Request for Intervention	
22	Scheduling Status	
23	Spoofed GPS	
24	Tanker Positioning Support	+Position +Speed +Docking Instructions

Figure 2: External Actor Item Flows to System

1.2.3 System Blackbox Item Flows

#	Part A	Item Flow	Part B
1	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Comms Subsystem	▶ Maintenance Request	ANGARS
2	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Autonomous Guidance & Navigation Sensors Subsystem	◀ Jamming ◀ Spoofed GPS	ANGARS
3	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Fuel Transfer Subsystem		ANGARS
4	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Autonomous Guidance & Navigation Sensors Subsystem	◀ Aircraft Status/Telemetry ▶ Tanker Positioning Support	ANGARS
5	ANGARS	▶ Operator Override	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Command & Control Subsystem
6	ANGARS	▶ Control Center Mission Commanding	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS::Comms Subsystem
7	ANGARS	▶ GPS Data	: 2.0 Logical Architecture::2.4 Logical Elements::ANGARS:: Autonomous Guidance & Navigatio...

Figure 3: System Blackbox Item Flow From-To

1.3 ANGARS Requirements Specification Updates

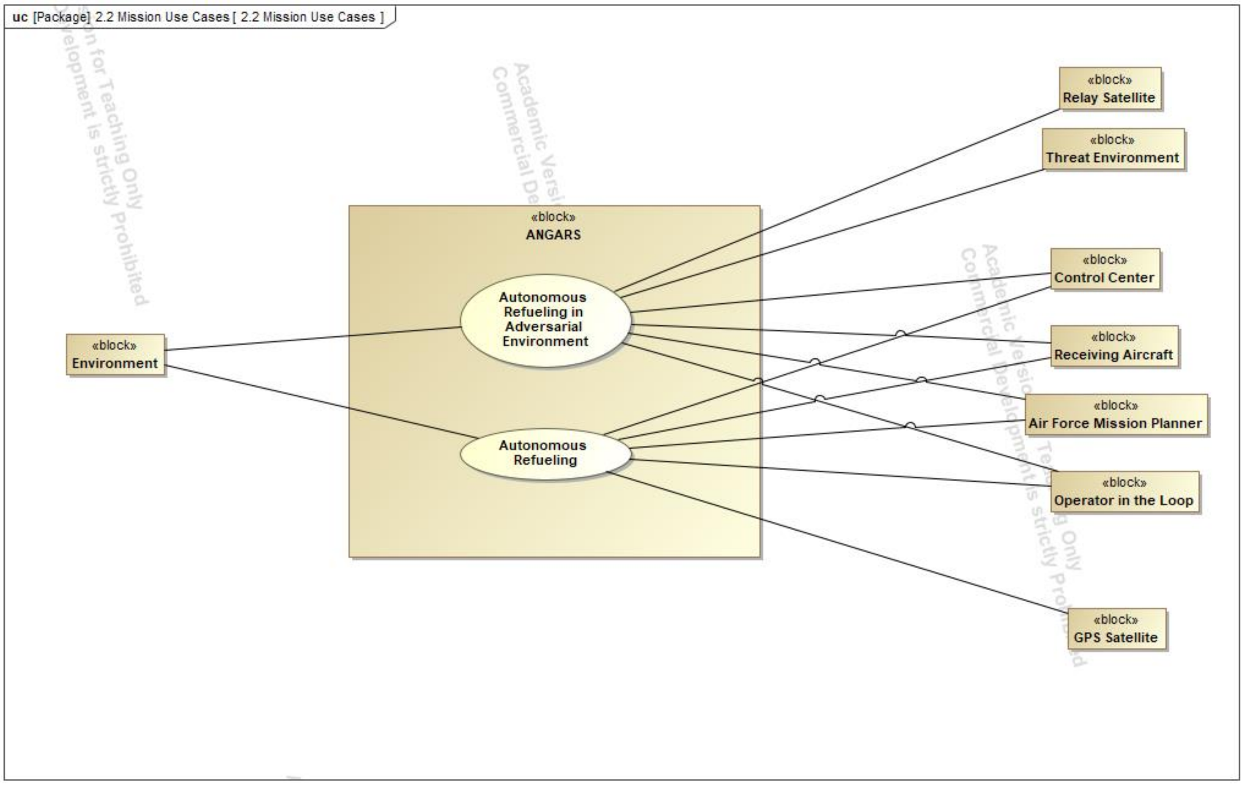
As systems engineering is an iterative process, requirements are continually being refined and updated to ultimately

Table 1: Requirements Tracker

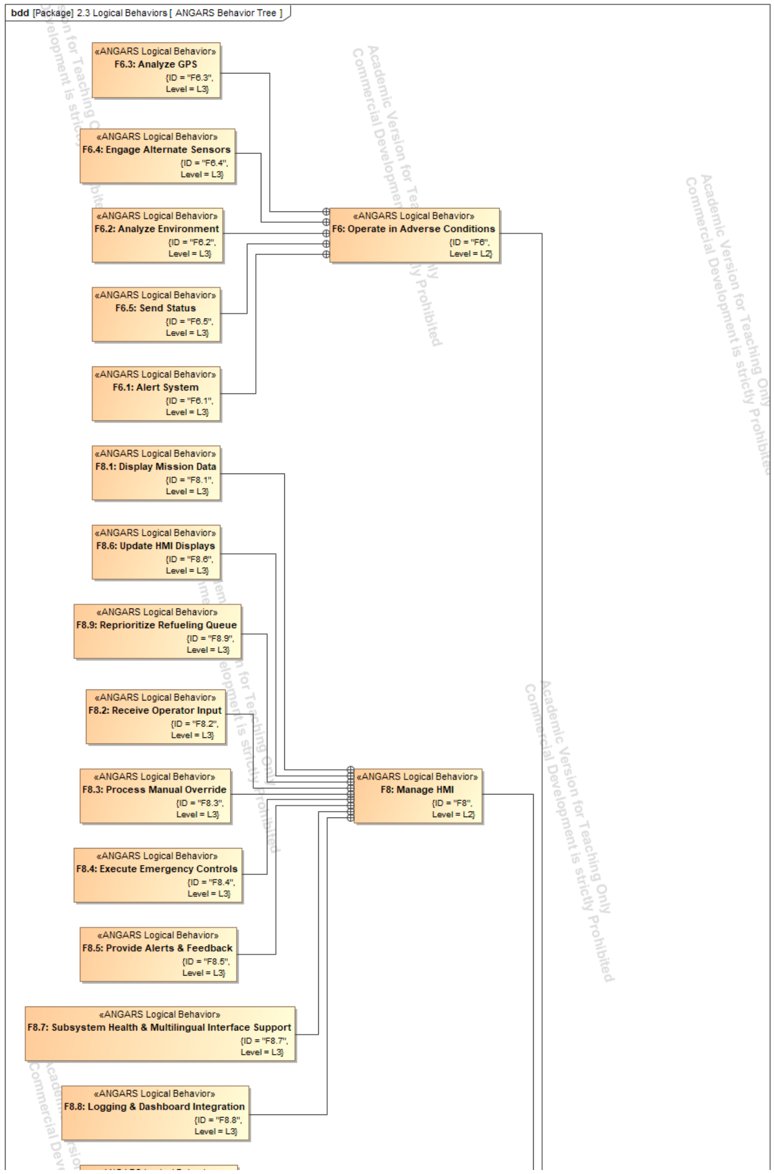
	Total	Quantitative	%	Binary	Qualitative
Requirements Analysis Report	154	48	31.1%	63	43
Functional Analysis Report	154	52	33.7%	63	39
Conceptual Design Report					
Trade Study					
Test Plan					
System Specifications					

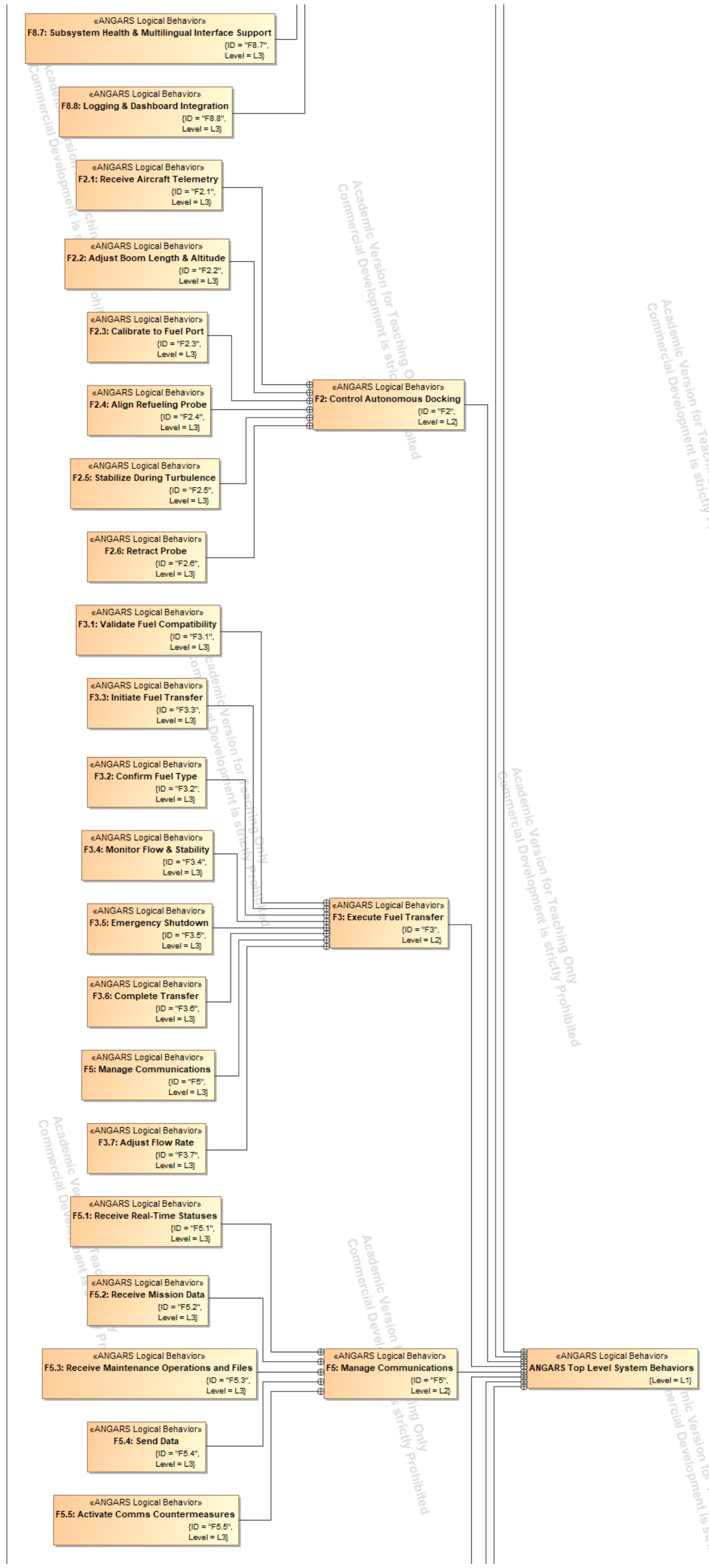
2 Functional Analysis

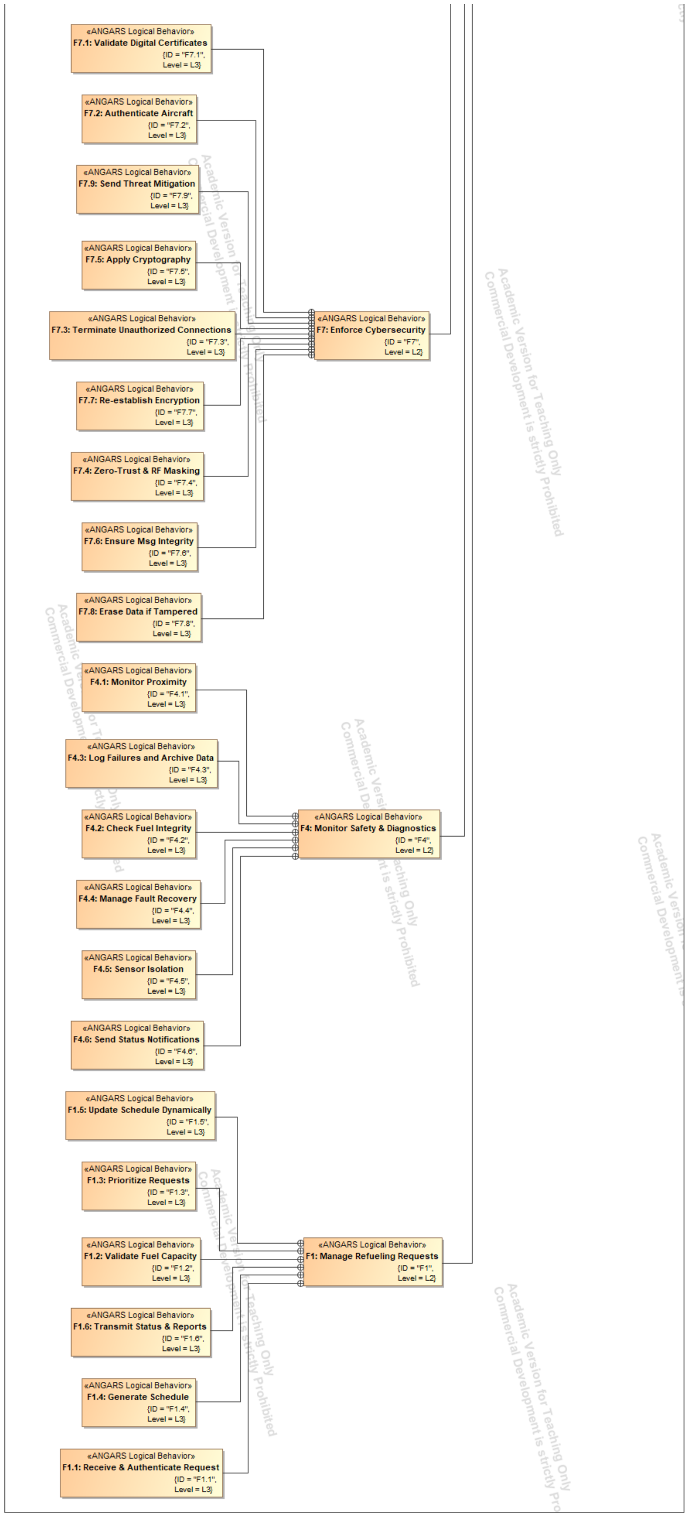
2.1 Use Case Diagram



2.2 Functional Tree




«ANGARS Logical Behavior»
F8.7: Subsystem Health & Multilingual Interface Support
{ID = "F8.7", Level = L3}
«ANGARS Logical Behavior»
F8.8: Logging & Dashboard Integration
{ID = "F8.8", Level = L3}
«ANGARS Logical Behavior»
F2.1: Receive Aircraft Telemetry
{ID = "F2.1", Level = L3}
«ANGARS Logical Behavior»
F2.2: Adjust Boom Length & Altitude
{ID = "F2.2", Level = L3}
«ANGARS Logical Behavior»
F2.3: Calibrate to Fuel Port
{ID = "F2.3", Level = L3}
«ANGARS Logical Behavior»
F2.4: Align Refueling Probe
{ID = "F2.4", Level = L3}
«ANGARS Logical Behavior»
F2.5: Stabilize During Turbulence
{ID = "F2.5", Level = L3}
«ANGARS Logical Behavior»
F2.6: Retract Probe
{ID = "F2.6", Level = L3}
«ANGARS Logical Behavior»
F2: Control Autonomous Docking
{ID = "F2", Level = L2}
«ANGARS Logical Behavior»
F3.1: Validate Fuel Compatibility
{ID = "F3.1", Level = L3}
«ANGARS Logical Behavior»
F3.3: Initiate Fuel Transfer
{ID = "F3.3", Level = L3}
«ANGARS Logical Behavior»
F3.2: Confirm Fuel Type
{ID = "F3.2", Level = L3}
«ANGARS Logical Behavior»
F3.4: Monitor Flow & Stability
{ID = "F3.4", Level = L3}
«ANGARS Logical Behavior»
F3.5: Emergency Shutdown
{ID = "F3.5", Level = L3}
«ANGARS Logical Behavior»
F3.6: Complete Transfer
{ID = "F3.6", Level = L3}
«ANGARS Logical Behavior»
F5: Manage Communications
{ID = "F5", Level = L3}
«ANGARS Logical Behavior»
F3.7: Adjust Flow Rate
{ID = "F3.7", Level = L3}
«ANGARS Logical Behavior»
F3: Execute Fuel Transfer
{ID = "F3", Level = L2}
«ANGARS Logical Behavior»
F5.1: Receive Real-Time Statuses
{ID = "F5.1", Level = L3}
«ANGARS Logical Behavior»
F5.2: Receive Mission Data
{ID = "F5.2", Level = L3}
«ANGARS Logical Behavior»
F5.3: Receive Maintenance Operations and Files
{ID = "F5.3", Level = L3}
«ANGARS Logical Behavior»
F5.4: Send Data
{ID = "F5.4", Level = L3}
«ANGARS Logical Behavior»
F5.5: Activate Comms Countermeasures
{ID = "F5.5", Level = L3}
«ANGARS Logical Behavior»
F5: Manage Communications
{ID = "F5", Level = L2}
«ANGARS Logical Behavior»
ANGARS Top Level System Behaviors
{Level = L1}
Academic Version for Teaching Only
Commercial Development is strictly Prohibited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«ANGARS Logical Behavior» F7.2: Authenticate Aircraft {ID = "F7.2", Level = L3}
«ANGARS Logical Behavior» F7.9: Send Threat Mitigation {ID = "F7.9", Level = L3}
«ANGARS Logical Behavior» F7.5: Apply Cryptography {ID = "F7.5", Level = L3}
«ANGARS Logical Behavior» F7.3: Terminate Unauthorized Connections {ID = "F7.3", Level = L3}
«ANGARS Logical Behavior» F7: Enforce Cybersecurity {ID = "F7", Level = L2}
«ANGARS Logical Behavior» F7.7: Re-establish Encryption {ID = "F7.7", Level = L3}
«ANGARS Logical Behavior» F7.4: Zero-Trust & RF Masking {ID = "F7.4", Level = L3}
«ANGARS Logical Behavior» F7.6: Ensure Msg Integrity {ID = "F7.6", Level = L3}
«ANGARS Logical Behavior» F7.8: Erase Data if Tampered {ID = "F7.8", Level = L3}
«ANGARS Logical Behavior» F4.1: Monitor Proximity {ID = "F4.1", Level = L3}
«ANGARS Logical Behavior» F4.3: Log Failures and Archive Data {ID = "F4.3", Level = L3}
«ANGARS Logical Behavior» F4.2: Check Fuel Integrity {ID = "F4.2", Level = L3}
«ANGARS Logical Behavior» F4: Monitor Safety & Diagnostics {ID = "F4", Level = L2}
«ANGARS Logical Behavior» F4.4: Manage Fault Recovery {ID = "F4.4", Level = L3}
«ANGARS Logical Behavior» F4.5: Sensor Isolation {ID = "F4.5", Level = L3}
«ANGARS Logical Behavior» F4.6: Send Status Notifications {ID = "F4.6", Level = L3}
«ANGARS Logical Behavior» F1.5: Update Schedule Dynamically {ID = "F1.5", Level = L3}
«ANGARS Logical Behavior» F1.3: Prioritize Requests {ID = "F1.3", Level = L3}
«ANGARS Logical Behavior» F1.2: Validate Fuel Capacity {ID = "F1.2", Level = L3}
«ANGARS Logical Behavior» F1: Manage Refueling Requests {ID = "F1", Level = L2}
«ANGARS Logical Behavior» F1.6: Transmit Status & Reports {ID = "F1.6", Level = L3}
«ANGARS Logical Behavior» F1.4: Generate Schedule {ID = "F1.4", Level = L3}
«ANGARS Logical Behavior» F1.1: Receive & Authenticate Request {ID = "F1.1", Level = L3}
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2.3 Functional Table

#	ID	Level	Name	Owner	Applied Stereotype
1		L1	ANGARS Top Level System Behaviors	ANGARS Top Level	ANGARS Logical Behavior [Class]
2	F1	L2	F1: Manage Refueling Requests	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
3	F2	L2	F2: Control Autonomous Docking	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
4	F3	L2	F3: Execute Fuel Transfer	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
5	F4	L2	F4: Monitor Safety & Diagnostics	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
6	F5	L2	F5: Manage Communications	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
7	F6	L2	F6: Operate in Adverse Conditions	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
8	F7	L2	F7: Enforce Cybersecurity	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
9	F8	L2	F8: Manage HMI	ANGARS Top Level System Behaviors	ANGARS Logical Behavior [Class]
10	F1.1	L3	F1.1: Receive & Authenticate Request	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
11	F1.2	L3	F1.2: Validate Fuel Capacity	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
12	F1.3	L3	F1.3: Prioritize Requests	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
13	F1.4	L3	F1.4: Generate Schedule	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
14	F1.5	L3	F1.5: Update Schedule Dynamically	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
15	F1.6	L3	F1.6: Transmit Status & Reports	F1: Manage Refueling Requests	ANGARS Logical Behavior [Class]
16	F2.1	L3	F2.1: Receive Aircraft Telemetry	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
17	F2.2	L3	F2.2: Adjust Boom Length & Altitude	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
18	F2.3	L3	F2.3: Calibrate to Fuel Port	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
19	F2.4	L3	F2.4: Align Refueling Probe	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
20	F2.5	L3	F2.5: Stabilize During Turbulence	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
21	F2.6	L3	F2.6: Retract Probe	F2: Control Autonomous Docking	ANGARS Logical Behavior [Class]
22	F3.1	L3	F3.1: Validate Fuel Compatibility	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
23	F3.2	L3	F3.2: Confirm Fuel Type	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
24	F3.3	L3	F3.3: Initiate Fuel Transfer	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
25	F3.4	L3	F3.4: Monitor Flow & Stability	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
26	F3.5	L3	F3.5: Emergency Shutdown	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
27	F3.6	L3	F3.6: Complete Transfer	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
28	F3.7	L3	F3.7: Adjust Flow Rate	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
29	F5	L3	F5: Manage Communications	F3: Execute Fuel Transfer	ANGARS Logical Behavior [Class]
30	F4.1	L3	F4.1: Monitor Proximity	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
31	F4.2	L3	F4.2: Check Fuel Integrity	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
32	F4.3	L3	F4.3: Log Failures and Archive Data	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
33	F4.4	L3	F4.4: Manage Fault Recovery	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
34	F4.5	L3	F4.5: Sensor Isolation	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
35	F4.6	L3	F4.6: Send Status Notifications	F4: Monitor Safety & Diagnostics	ANGARS Logical Behavior [Class]
36	F5.1	L3	F5.1: Receive Real-Time Statuses	F5: Manage Communications	ANGARS Logical Behavior [Class]
37	F5.2	L3	F5.2: Receive Mission Data	F5: Manage Communications	ANGARS Logical Behavior [Class]
38	F5.3	L3	F5.3: Receive Maintenance Operations and Fil	F5: Manage Communications	ANGARS Logical Behavior [Class]
39	F5.4	L3	F5.4: Send Data	F5: Manage Communications	ANGARS Logical Behavior [Class]
40	F5.5	L3	F5.5: Activate Comms Countermeasures	F5: Manage Communications	ANGARS Logical Behavior [Class]
41	F6.1	L3	F6.1: Alert System	F6: Operate in Adverse Conditions	ANGARS Logical Behavior [Class]
42	F6.2	L3	F6.2: Analyze Environment	F6: Operate in Adverse Conditions	ANGARS Logical Behavior [Class]
43	F6.3	L3	F6.3: Analyze GPS	F6: Operate in Adverse Conditions	ANGARS Logical Behavior [Class]
44	F6.4	L3	F6.4: Engage Alternate Sensors	F6: Operate in Adverse Conditions	ANGARS Logical Behavior [Class]
45	F6.5	L3	F6.5: Send Status	F6: Operate in Adverse Conditions	ANGARS Logical Behavior [Class]
46	F7.1	L3	F7.1: Validate Digital Certificates	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
47	F7.2	L3	F7.2: Authenticate Aircraft	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
48	F7.3	L3	F7.3: Terminate Unauthorized Connections	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
49	F7.4	L3	F7.4: Zero-Trust & RF Masking	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
50	F7.5	L3	F7.5: Apply Cryptography	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
51	F7.6	L3	F7.6: Ensure Msg Integrity	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
52	F7.7	L3	F7.7: Re-establish Encryption	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
53	F7.8	L3	F7.8: Erase Data if Tampered	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
54	F7.9	L3	F7.9: Send Threat Mitigation	F7: Enforce Cybersecurity	ANGARS Logical Behavior [Class]
55	F8.1	L3	F8.1: Display Mission Data	F8: Manage HMI	ANGARS Logical Behavior [Class]
56	F8.2	L3	F8.2: Receive Operator Input	F8: Manage HMI	ANGARS Logical Behavior [Class]
57	F8.3	L3	F8.3: Process Manual Override	F8: Manage HMI	ANGARS Logical Behavior [Class]
58	F8.4	L3	F8.4: Execute Emergency Controls	F8: Manage HMI	ANGARS Logical Behavior [Class]
59	F8.5	L3	F8.5: Provide Alerts & Feedback	F8: Manage HMI	ANGARS Logical Behavior [Class]
60	F8.6	L3	F8.6: Update HMI Displays	F8: Manage HMI	ANGARS Logical Behavior [Class]
61	F8.7	L3	F8.7: Subsystem Health & Multilingual Interfa	F8: Manage HMI	ANGARS Logical Behavior [Class]
62	F8.8	L3	F8.8: Logging & Dashboard Integration	F8: Manage HMI	ANGARS Logical Behavior [Class]
63	F8.9	L3	F8.9: Reprioritize Refueling Queue	F8: Manage HMI	ANGARS Logical Behavior [Class]

2.4 Functional Description Diagrams

2.4.1 L1 Top Level ANGARS Behaviors

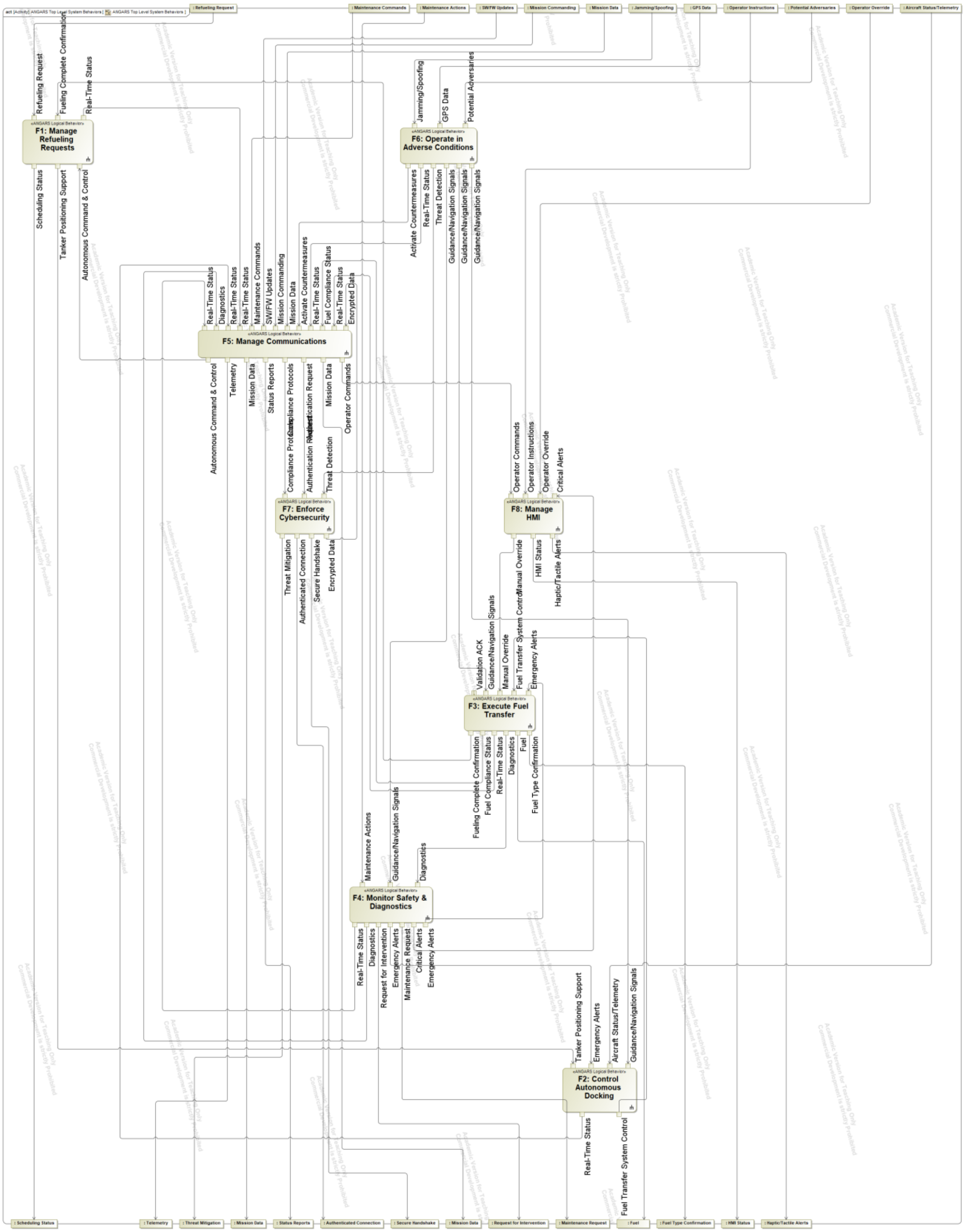


Figure 4: Top-Level ANGARS System Behavior

2.4.2 L2 ANGARS Behaviors

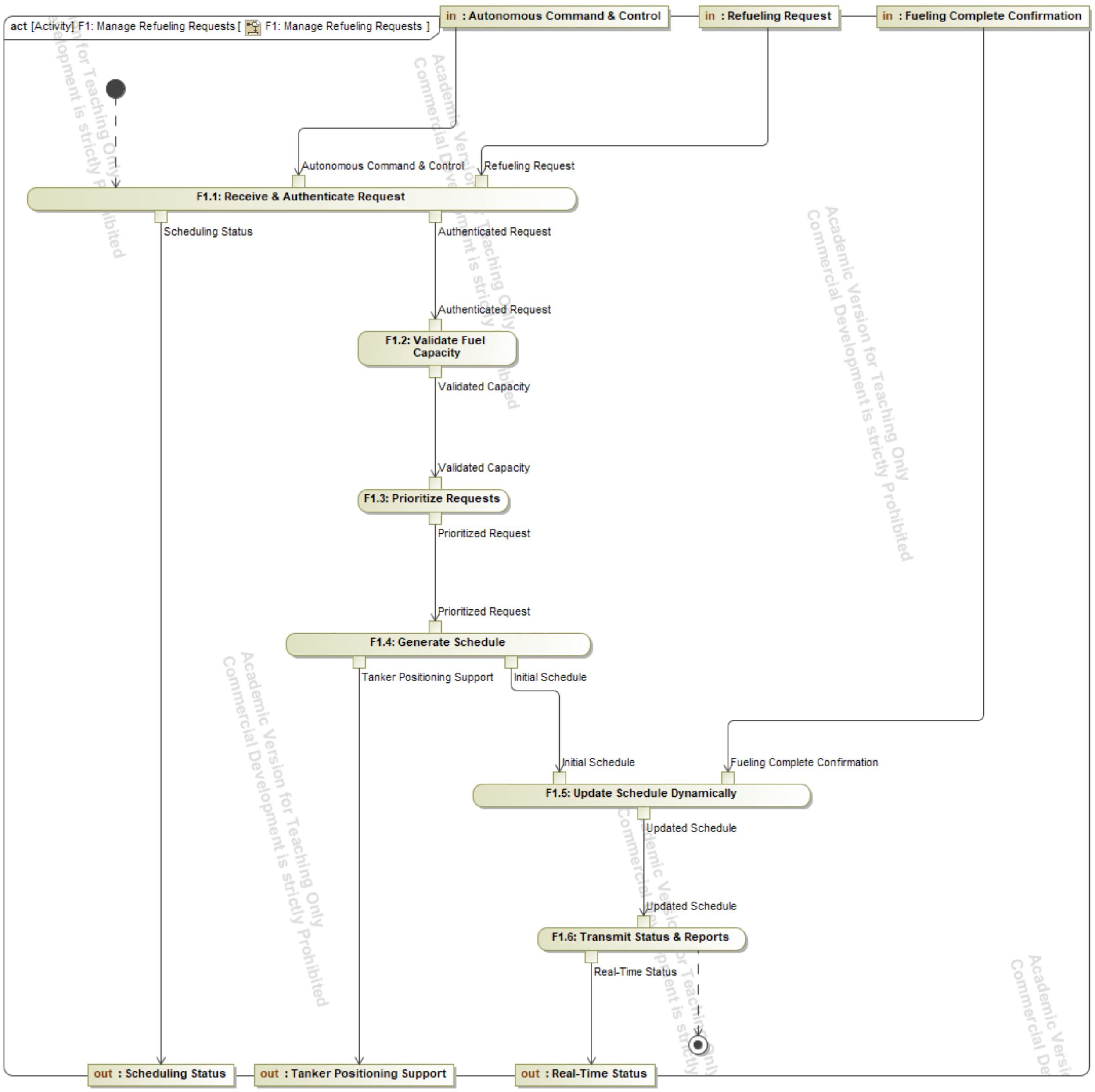


Figure 5: F1: Manage Refueling Requests

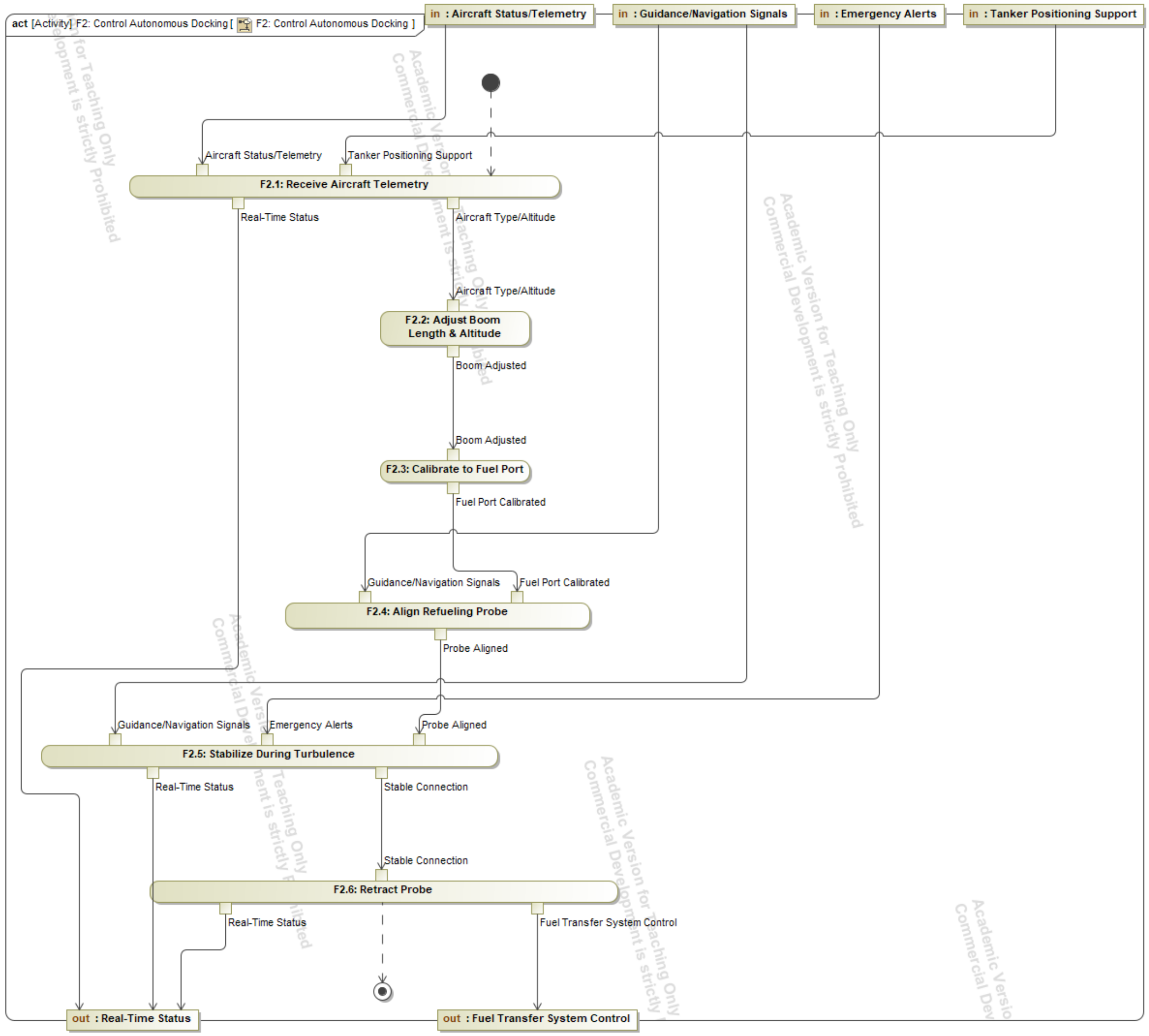


Figure 6: F2: Control Autonomous Docking

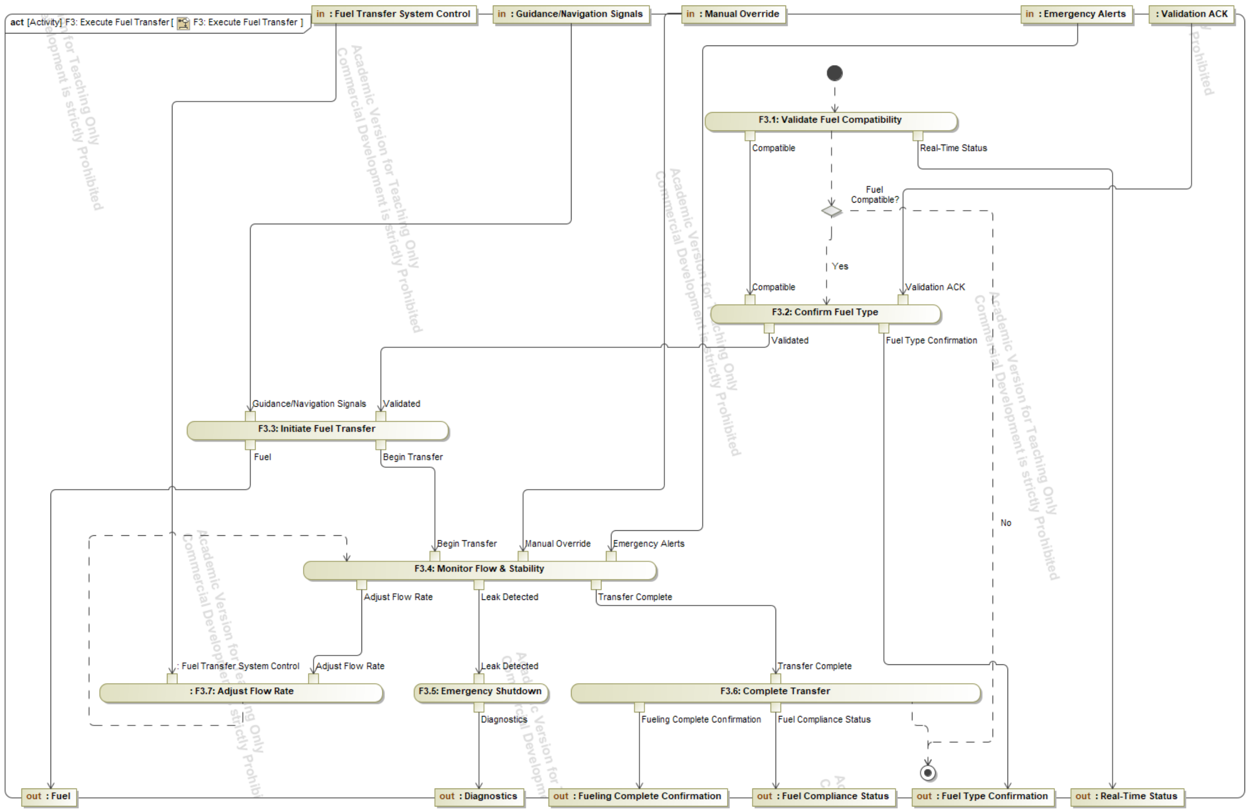


Figure 7: F3: Execute Fuel Transfer

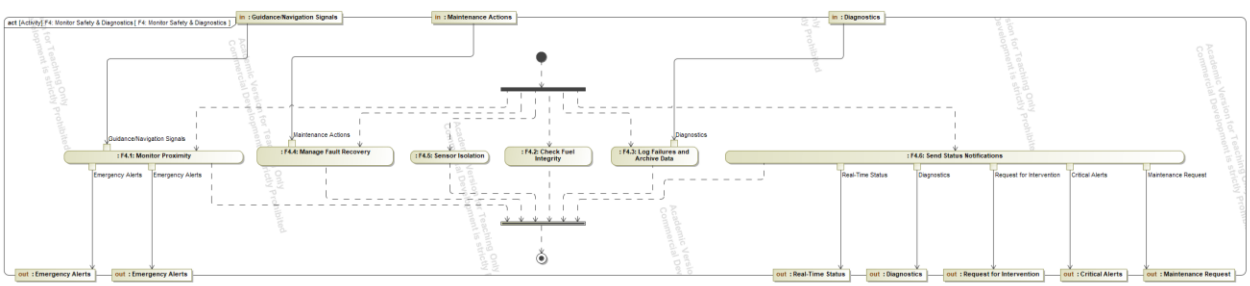


Figure 8: F4: Monitor Safety & Diagnostics

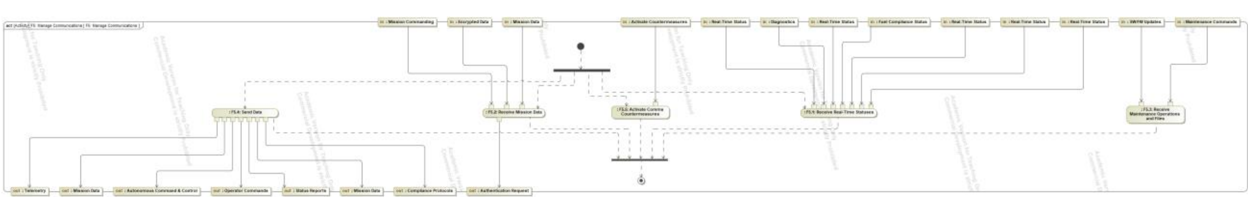


Figure 9: F5: Manage Communications

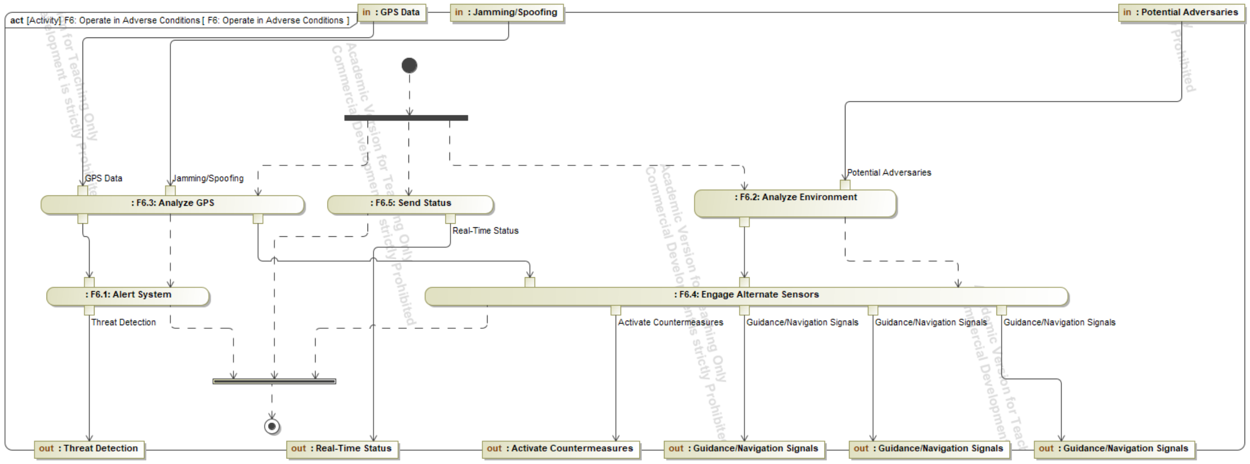


Figure 10: F6: Operate in Adverse Conditions

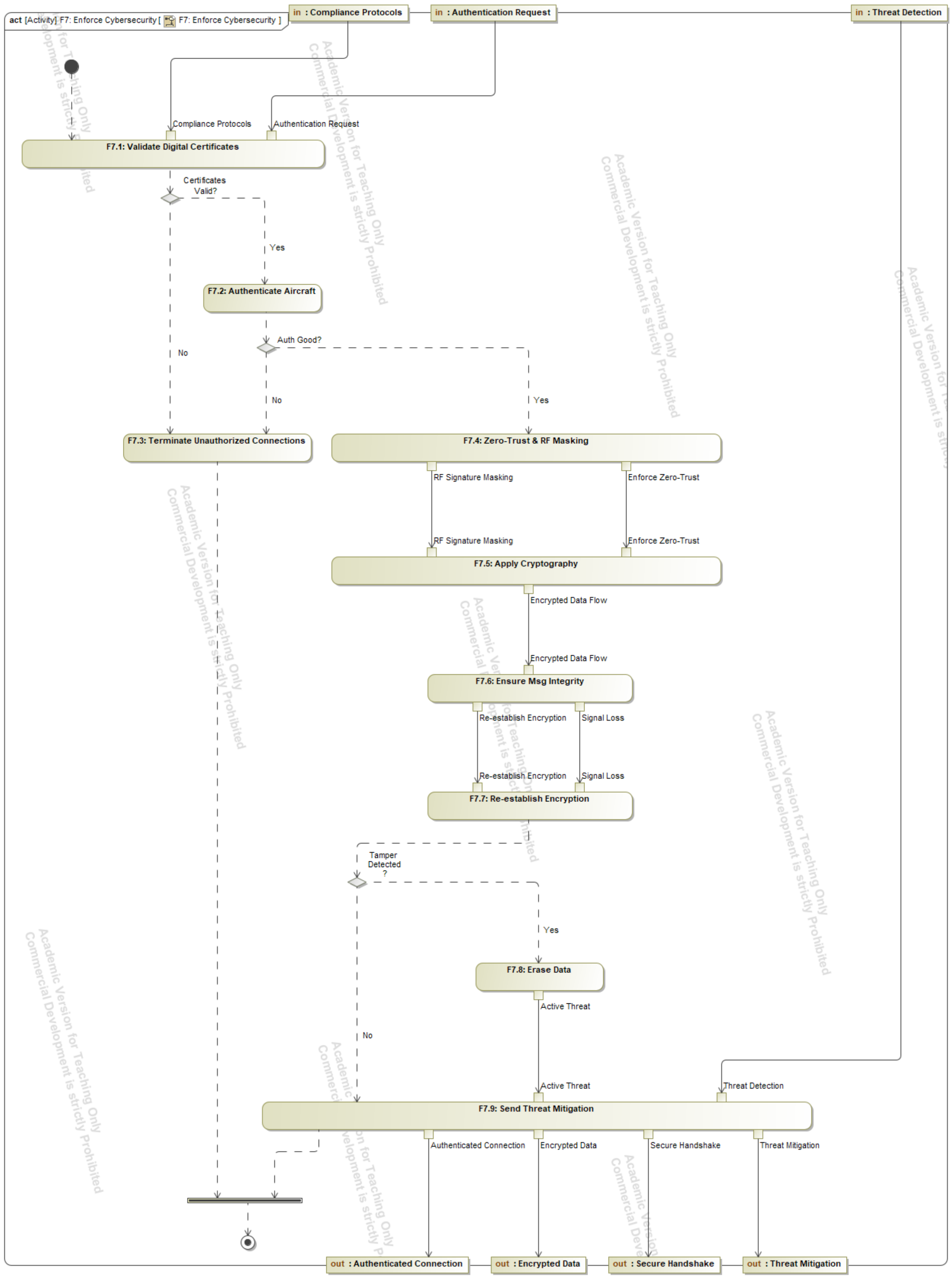


Figure 11: F7: Enforce Cybersecurity

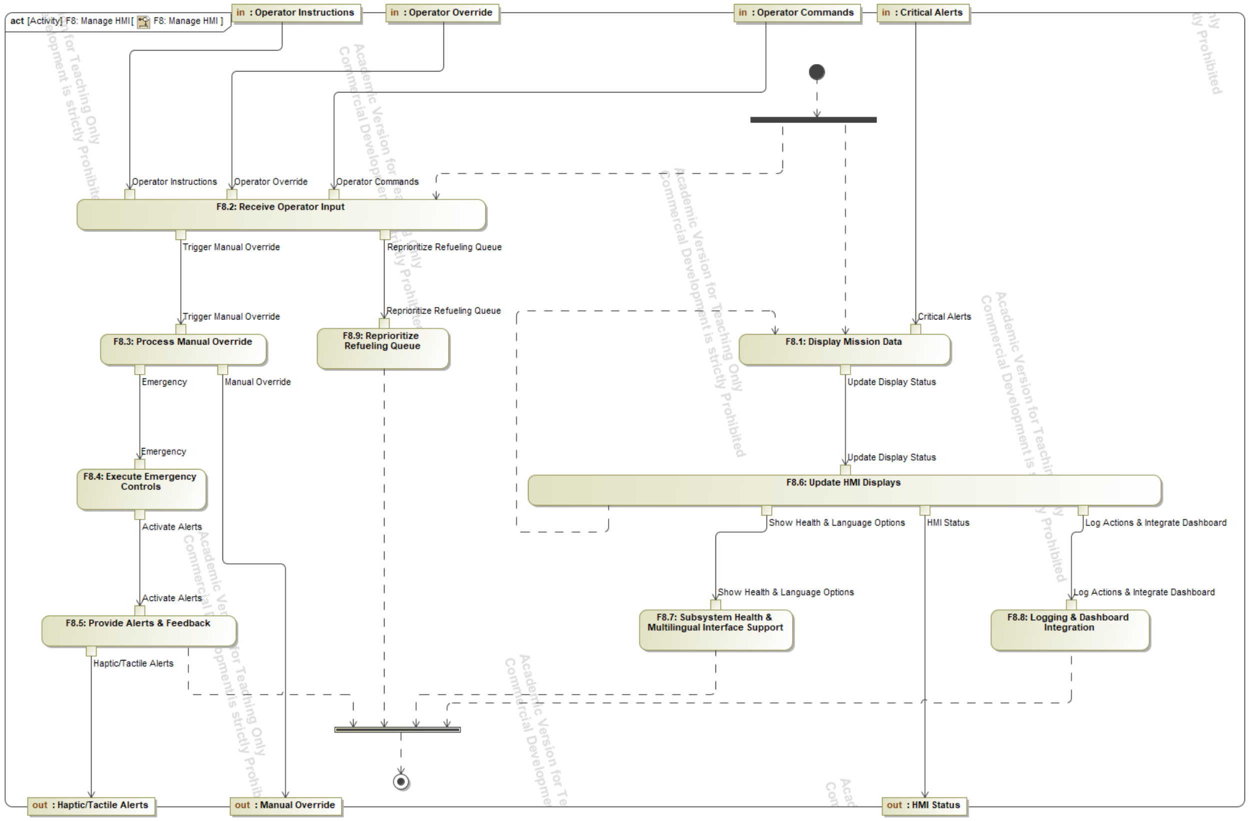


Figure 12: F8: Manage HMI

2.5 Functional N2

2.5.1 Internal Function-Function N2 Interactions Diagram

Sender /Receiver	F1	F2	F3	F4	F5	F6	F7	F8
F1: Manage Refueling Requests		Tanker Positioning Support	-	-	Real-Time Status	-	-	-
F2: Control Autonomous Docking	-		Fuel Transfer System Control	-	Real-Time Status	-	-	-
F3: Execute Fuel Transfer	Fueling Complete Confirmation	-		Diagnostics	Fuel Compliance Status Real-Time Status	-	-	-
F4: Monitor Safety & Diagnostics	-	Emergency Alerts	Emergency Alerts		Diagnostics Real-Time Status	-	-	Critical Alerts
F5: Manage Communications	Autonomous Command & Control	-	-	-		-	Authentication Request Compliance Protocols	Operator Commands
F6: Operate in Adverse Conditions	-	Guidance/Navigation Signals	Guidance/Navigation Signals	Guidance/Navigation Signals	Real-Time Status Activate Countermeasures		Threat Detection	-
F7: Enforce Cybersecurity	-	-	-	-	Encrypted Data	-		-
F8: Manage HMI	-	-	Manual Override	-	-	-	-	

Figure 13: Internal Function-Function N2

2.5.2 External Functional-Actor N2 Interactions Diagram

Actor / Function	F1: Manage Refueling Requests	F2: Control Autonomous Docking	F3: Execute Fuel Transfer	F4: Monitor Safety & Diagnostics	F5: Manage Communications	F6: Operate in Adverse Conditions	F7: Enforce Cybersecurity	F8: Manage HMI
1. Air Force Mission Planner	-	-	-	-	Send: Mission Commanding Receive: Mission Data	-	-	-
2. Control Center	-	-	-	-	Receive: Status Reports, Telemetry, Mission Data	-	Receive: Threat Mitigation	-
3. Environment	-	-	-	-	-	-	-	-
4. GPS Satellite	-	-	-	-	-	Send: GPS Data	-	-
5. Maintainer	-	-	-	Send: Maintenance Actions Receive: Maintenance Request Request for Intervention	Send: Maintenance Commands, SW/FW Updates	-	-	-
6. Operator in the Loop	-	-	-	-	Receive: Operator Commands	-	-	Send: Operator Instructions & Operator Override Receive: Haptic/Tactile Alerts, HMI Status
7. Receiving Aircraft	Send: Refueling Request Receive: Scheduling Status	Send: Aircraft Status/Telemetry	Receive: Fuel Fuel Type Confirmation	-	-	-	-	-
8. Relay Satellite	-	-	-	-	-	-	Receive: Authenticated Connection Secure Handshake	-
9. Threat Environment	-	-	-	-	-	Send: Jamming/Spoofin Potential Adversaries	-	-

Figure 14: External Actor-Function N2

2.6 States and Modes Diagram

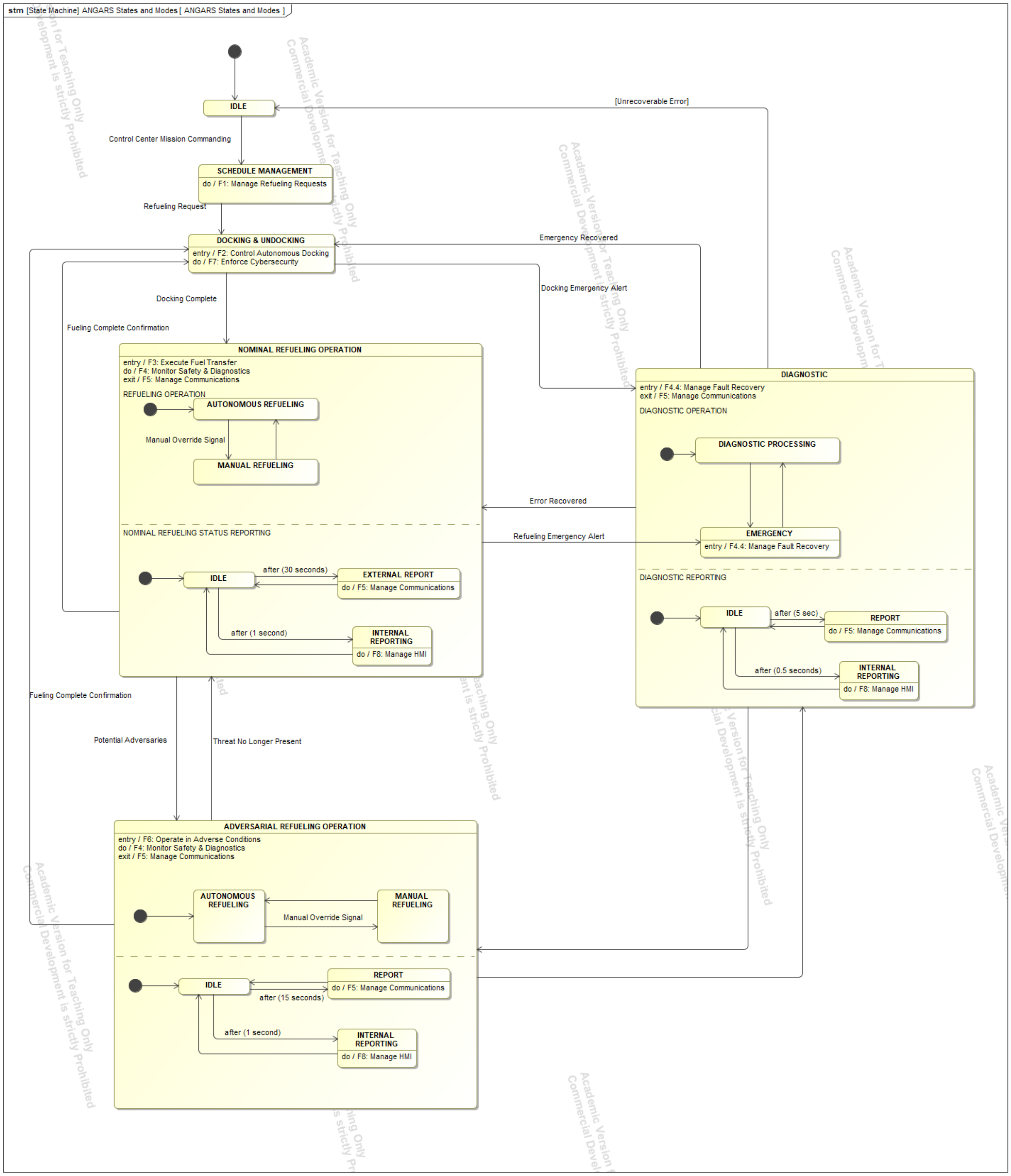
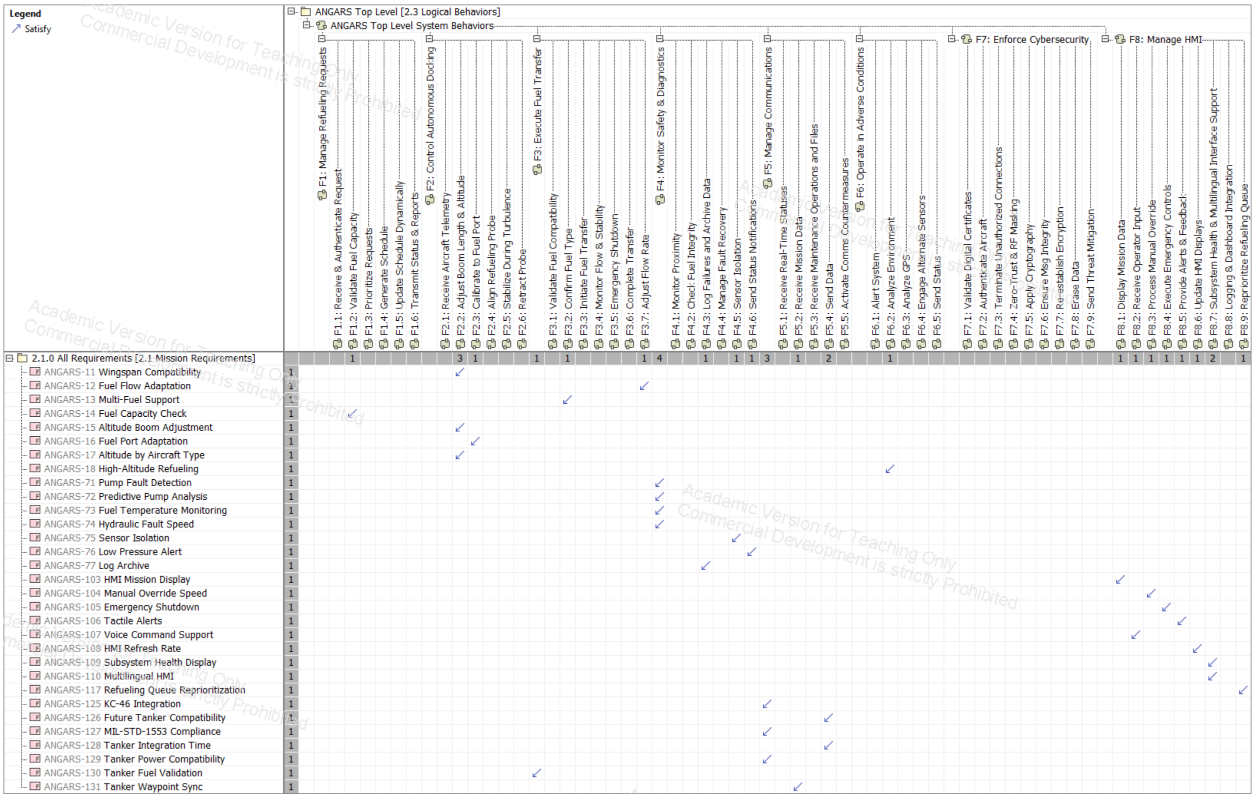


Figure 15: ANGARS States

3 System Traceability

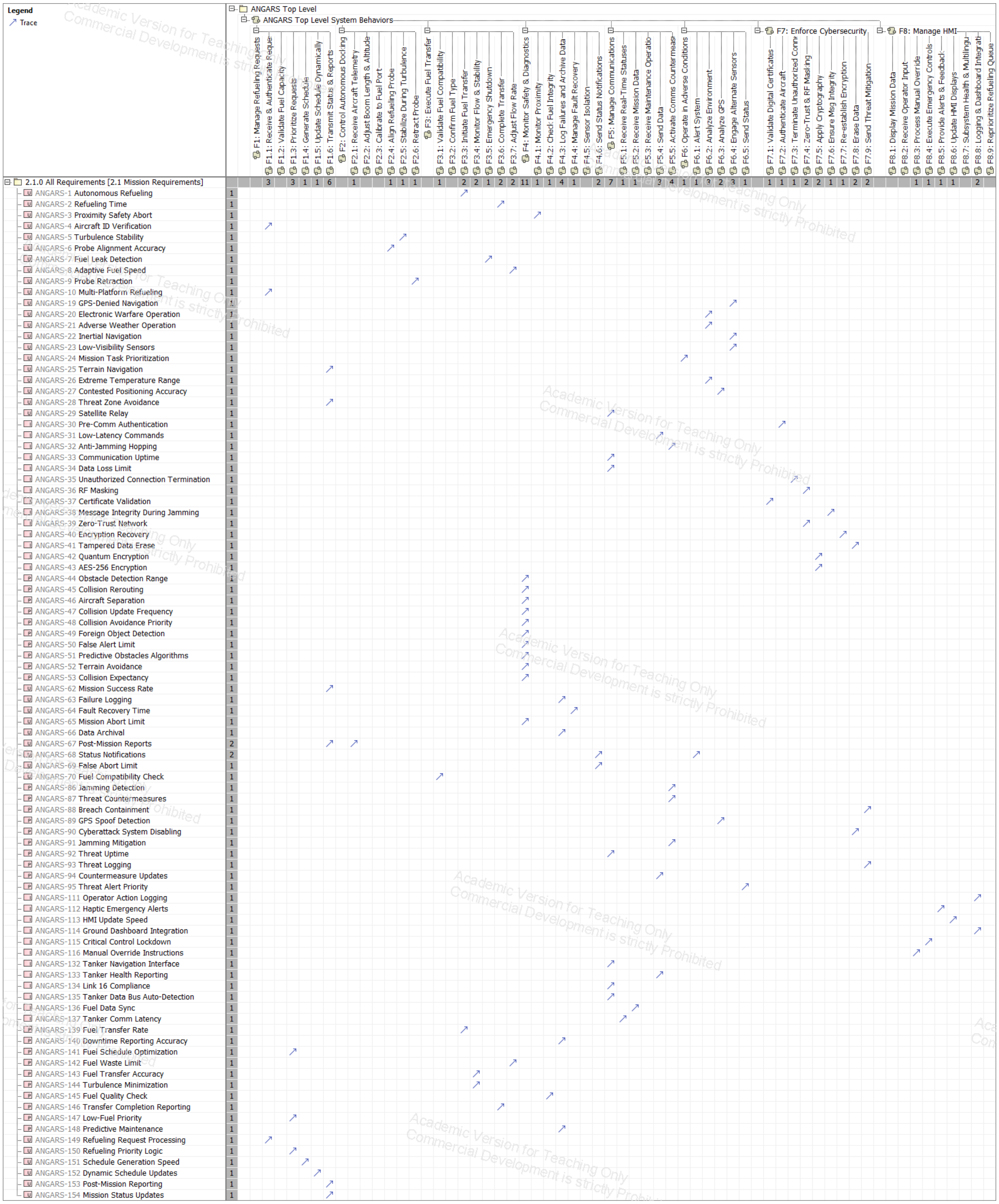
3.1 Functional Requirement Satisfy Relationships

Functional Requirement can be satisfied by system functionality. The matrix below shows which requirements are satisfied by system functionality.



3.2 Non-Functional Requirement Traceability Relationships

Non-Functional Requirements cannot be satisfied by system functionality, however, traceability can be allocated between requirements and functionality to support satisfaction traceability via model querying when that information is defined.


Legend
Trace
ANGARS Top Level
ANGARS Top Level System Behaviors
F1: Manage Refueling Requests
F1.1: Receive & Authenticate Reque
F1.2: Validate Fuel Capacity
F1.3: Prioritize Requests
F1.4: Generate Schedule
F1.5: Update Schedule Dynamically
F1.6: Transmit Status & Reports
F2: Control Autonomous Docking
F2.1: Receive Aircraft Telemetry
F2.2: Adjust Boom Length & Altitude
F2.3: Calibrate to Fuel Port
F2.4: Align Refueling Probe
F2.5: Stabilize During Turbulence
F2.6: Retract Probe
F3: Execute Fuel Transfer
F3.1: Validate Fuel Compatibility
F3.2: Confirm Fuel Type
F3.3: Initiate Fuel Transfer
F3.4: Monitor Flow & Stability
F3.5: Emergency Shutdown
F3.6: Complete Transfer
F3.7: Adjust Flow Rate
F4: Monitor Safety & Diagnostics
F4.1: Monitor Proximity
F4.2: Check Fuel Integrity
F4.3: Log Failures and Archive Data
F4.4: Manage Fault Recovery
F4.5: Sensor Isolation
F4.6: Send Status Notifications
F5: Manage Communications
F5.1: Receive Real-Time Statuses
F5.2: Receive Mission Data
F5.3: Receive Maintenance Operatio
F5.4: Send Data
F5.5: Activate Comms Countermea
F6: Operate in Adverse Conditions
F6.1: Alert System
F6.2: Analyze Environment
F6.3: Analyze GPS
F6.4: Engage Alternate Sensors
F6.5: Send Status
F7: Enforce Cybersecurity
F7.1: Validate Digital Certificates
F7.2: Authenticate Aircraft
F7.3: Terminate Unauthorized Conn
F7.4: Zero-Trust & RF Masking
F7.5: Apply Cryptography
F7.6: Ensure Msg Integrity
F7.7: Re-establish Encryption
F7.8: Erase Data
F7.9: Send Threat Mitigation
F8: Manage HMI
F8.1: Display Mission Data
F8.2: Receive Operator Input
F8.3: Process Manual Override
F8.4: Execute Emergency Controls
F8.5: Provide Alerts & Feedback
F8.6: Update HMI Displays
F8.7: Subsystem Health & Multilingu
F8.8: Logging & Dashboard Integrat
F8.9: Reprioritize Refueling Queue
2.1.0 All Requirements [2.1 Mission Requirements]
ANGARS-1 Autonomous Refueling
ANGARS-2 Refueling Time
ANGARS-3 Proximity Safety Abort
ANGARS-4 Aircraft ID Verification
ANGARS-5 Turbulence Stability
ANGARS-6 Probe Alignment Accuracy
ANGARS-7 Fuel Leak Detection
ANGARS-8 Adaptive Fuel Speed
ANGARS-9 Probe Retraction
ANGARS-10 Multi-Platform Refueling
ANGARS-19 GPS-Denied Navigation
ANGARS-20 Electronic Warfare Operation
ANGARS-21 Adverse Weather Operation
ANGARS-22 Inertial Navigation
ANGARS-23 Low-Visibility Sensors
ANGARS-24 Mission Task Prioritization
ANGARS-25 Terrain Navigation
ANGARS-26 Extreme Temperature Range
ANGARS-27 Contested Positioning Accuracy
ANGARS-28 Threat Zone Avoidance
ANGARS-29 Satellite Relay
ANGARS-30 Pre-Comm Authentication
ANGARS-31 Low-Latency Commands
ANGARS-32 Anti-Jamming Hopping
ANGARS-33 Communication Uptime
ANGARS-34 Data Loss Limit
ANGARS-35 Unauthorized Connection Termination
ANGARS-36 RF Masking
ANGARS-37 Certificate Validation
ANGARS-38 Message Integrity During Jamming
ANGARS-39 Zero-Trust Network
ANGARS-40 Encryption Recovery
ANGARS-41 Tampered Data Erase
ANGARS-42 Quantum Encryption
ANGARS-43 AES-256 Encryption
ANGARS-44 Obstacle Detection Range
ANGARS-45 Collision Rerouting
ANGARS-46 Aircraft Separation
ANGARS-47 Collision Update Frequency
ANGARS-48 Collision Avoidance Priority
ANGARS-49 Foreign Object Detection
ANGARS-50 False Alert Limit
ANGARS-51 Predictive Obstacles Algorithms
ANGARS-52 Terrain Avoidance
ANGARS-53 Collision Expectancy
ANGARS-62 Mission Success Rate
ANGARS-63 Failure Logging
ANGARS-64 Fault Recovery Time
ANGARS-65 Mission Abort Limit
ANGARS-66 Data Archival
ANGARS-67 Post-Mission Reports
ANGARS-68 Status Notifications
ANGARS-69 False Abort Limit
ANGARS-70 Fuel Compatibility Check
ANGARS-86 Jamming Detection
ANGARS-87 Threat Countermeasures
ANGARS-88 Breach Containment
ANGARS-89 GPS Spoof Detection
ANGARS-90 Cyberattack System Disabling
ANGARS-91 Jamming Mitigation
ANGARS-92 Threat Uptime
ANGARS-93 Threat Logging
ANGARS-94 Countermeasure Updates
ANGARS-95 Threat Alert Priority
ANGARS-111 Operator Action Logging
ANGARS-112 Haptic Emergency Alerts
ANGARS-113 HMI Update Speed
ANGARS-114 Ground Dashboard Integration
ANGARS-115 Critical Control Lockdown
ANGARS-116 Manual Override Instructions
ANGARS-132 Tanker Navigation Interface
ANGARS-133 Tanker Health Reporting
ANGARS-134 Link 16 Compliance
ANGARS-135 Tanker Data Bus Auto-Detection
ANGARS-136 Fuel Data Sync
ANGARS-137 Tanker Comm Latency
ANGARS-139 Fuel Transfer Rate
ANGARS-140 Downtime Reporting Accuracy
ANGARS-141 Fuel Schedule Optimization
ANGARS-142 Fuel Waste Limit
ANGARS-143 Fuel Transfer Accuracy
ANGARS-144 Turbulence Minimization
ANGARS-145 Fuel Quality Check
ANGARS-146 Transfer Completion Reporting
ANGARS-147 Low-Fuel Priority
ANGARS-148 Predictive Maintenance
ANGARS-149 Refueling Request Processing
ANGARS-150 Refueling Priority Logic
ANGARS-151 Schedule Generation Speed
ANGARS-152 Dynamic Schedule Updates
ANGARS-153 Post-Mission Reporting
ANGARS-154 Mission Status Updates
Academic Version for Teaching Only
Commercial Development is strictly Prohibited

4 Model Structure and Updates

4.1 Common Modeling Profile

A common modeling profile will be utilized throughout this Capstone. This modeling profile is being leveraged from prior Johns Hopkins modeling classes. This common modeling profile will be applied when necessary to modeling elements to facilitate traceability and model querying to better support needs of Capstone expectations as the semester progresses.

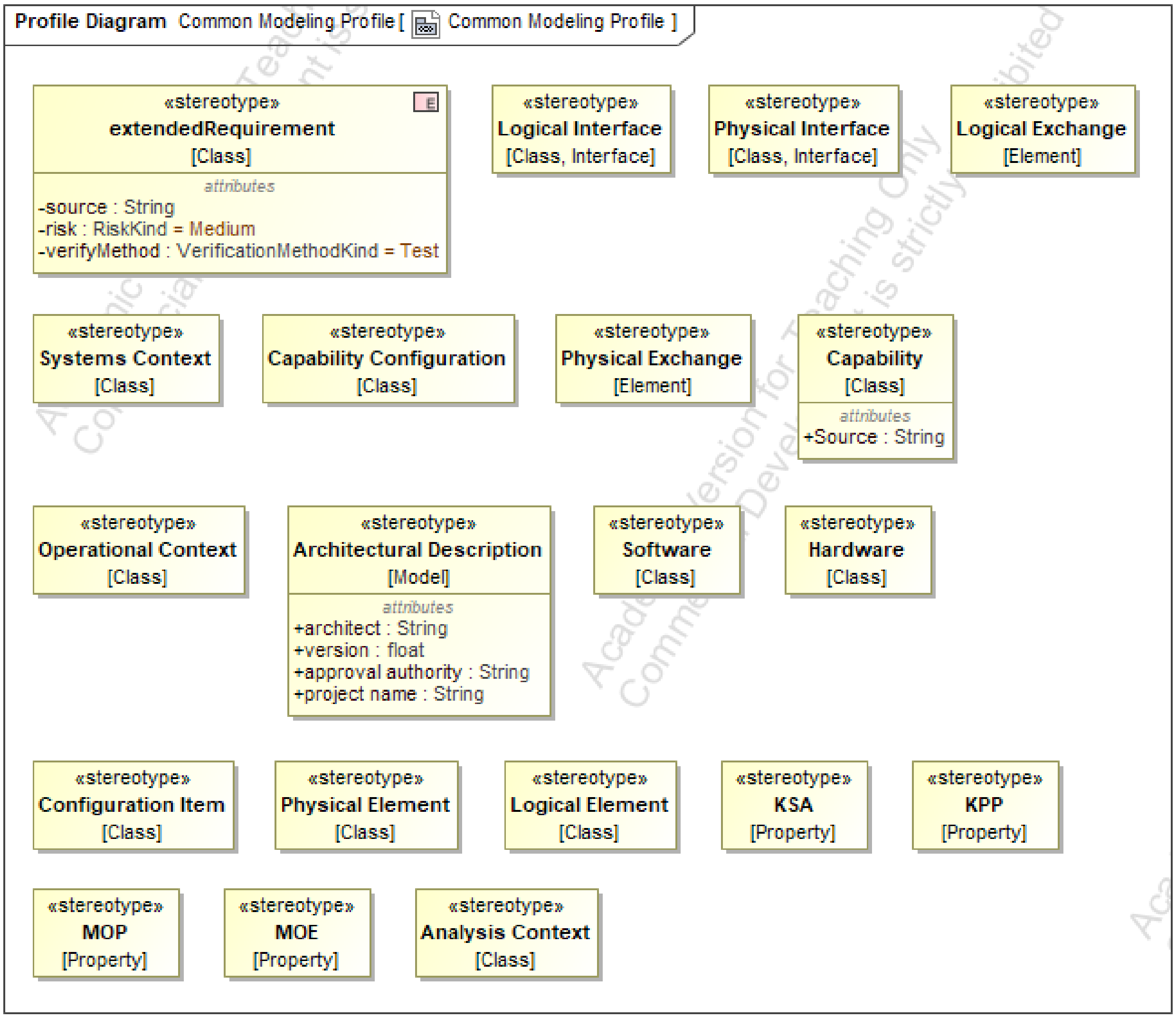


Figure 16: Common Modeling Profile

4.2 ANGARS Modeling Profile

The below modeling profile is in development and is being used to support custom modeling stereotypes to facilitate better tracking of important system elements. Current development includes;

- ANGARS Requirements stereotype – facilitates storing of requirement classification designation of Quantitative, Binary, Qualitative and a binary Key Performance Parameter (KPP) to better support Capstone requirements of tracking these attributes within requirements.
- ANGARS Logical Behavior – similar to requirements, Capstone class requirements drive a need to store additional contextual information along with behaviors. Current model design provides custom ID and Leveling to show requirements hierarchy.

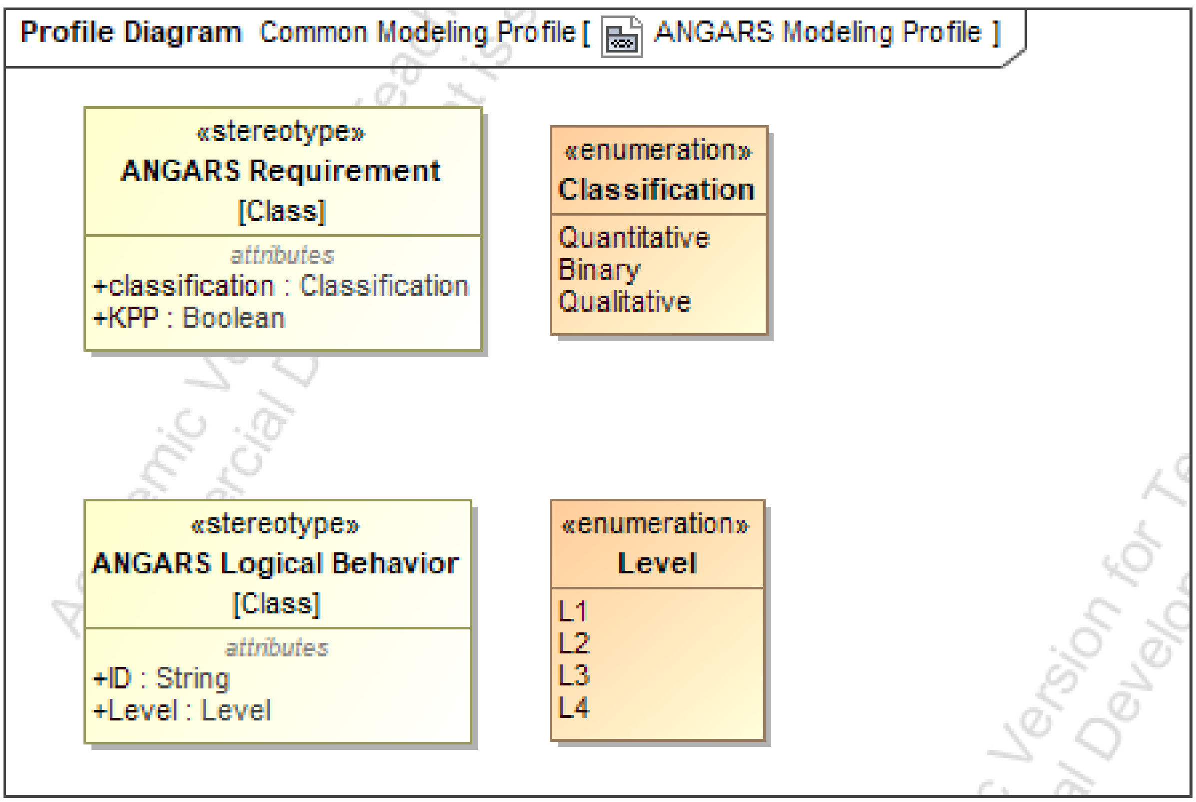


Figure 17: Custom ANGARS Modeling Profile

Appendix A – Schedule and EVM Updates

Schedule

RAR and CONOPS has not received approval at the time of this reports submittal and as such WBS cannot be updated to reflect actual completion information.

Table 2: Capstone Schedule

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
1	**Project Concept & Proposal**	50	10-Dec	21-Jan	25	10-Dec	1/19/2025
1.1	Intro	5					
1.3	System Need	5					
1.4	System Description	15					
1.5	Background	10					
1.6	Justification	5					
1.7	Draft Cleanup & Submittal	5					
1.8	Final Revisions & Submittal	5					
2	**Requirements Analysis Report (RAR) & CONOPS**	45	21-Jan	2-Feb			
2.1	Requirements Definition & Description	8					
2.2	Requirements Verification Plan	6					
2.3	User/Stakeholder Needs Anaylsis & Report	7					
2.4	Key Performance Parameters (KPP) Identification & Compilation	6					
2.5	CONOPS - Concept Diagram	6					
2.6	CONOPS - Potential Users/Operators List	4					
2.7	CONOPS - Use Case Scenario Definition	5					
2.8	RAR & CONOPS Cleanup & Submittal	3					
3	**Functional Analysis Report (FAR)**	37	2-Feb	16-Feb			
3.1	Functional Tree Definition	5					
3.2	Functional Decomp / Tree	5					
3.3	Context Diagram UPDATES	3					
3.4	Function List Development	4					
3.5	Functional Flow Diagrams Creation	6					
3.6	N2 Diagrams Development	5					
3.7	Requirements-Functions Traceability Matrix	4					
3.8	System States & Modes Definition	3					
3.9	FAR Cleanup & Submittal	2					
4	**Trade Study**	32	16-Feb	23-Feb			
4.1	Trade Study Purpose Definition	3					
4.2	Selection Criteria Development	4					
4.3	Alternatives Analysis	5					
4.4	Criteria Weights Development	4					
4.5	Utility Functions Development	4					
4.6	Raw Criteria Values Collection	4					
4.7	Weighted Utility Calculations	3					
4.8	Sensitivity Analysis	3					
4.9	Trade Study Cleanup & Submittal	2					

WBS ID	Task Name	Planned			Actual		
		Work Hours	Milestone Start	Milestone Finish	Work Hours	Milestone Start	Milestone Finish
5	**Concept Design Report**	37	23-Feb	2-Mar			
5.1	Physical Context Diagram Development	5					
5.2	Physical Component Tree Creation	4					
5.3	Subsystem Descriptions	5					
5.4	Physical Block Diagrams Development	6					
5.5	Physical N2 Diagrams Creation	5					
5.6	Data Flow Diagrams	4					
5.7	Interface Definitions & Descriptions	3					
5.8	Functions-Components Mapping	3					
5.9	Concept Design Cleanup & Submittal	2					
6	**Test Plan**	27	2-Mar	9-Mar			
6.1	Test Objectives Definition	3					
6.2	Requirements-Test Objectives Traceability	4					
6.3	Test Environment Description	3					
6.4	Test Equipment Identification	3					
6.5	Test Subjects & Articles Definition	3					
6.6	Success Criteria Development	3					
6.7	Metrics Definition	3					
6.8	Test Execution Planning	3					
6.9	Test Plan Cleanup & Submittal	2					
7	**A-Specification Report**	22	9-Mar	16-Mar			
7.1	Requirements Update & Organization	4					
7.2	KPP Updates	3					
7.3	Support Requirements Development	3					
7.4	Safety Requirements Development	3					
7.5	HSI Requirements Development	4					
7.6	Requirements-Functions Traceability Update	3					
7.7	A-Spec Cleanup & Submittal	2					
8	**Risk Management Report**	15	16-Mar	30-Mar			
8.1	Risk Definitions Development	2					
8.2	Risk Categorization	2					
8.3	Individual Risk Assessments	3					
8.4	Mitigation Steps Documentation	3					
8.5	Risk Progression Analysis	3					
8.6	Risk Report Cleanup & Submittal	2					
9	**Draft Final Report**	16	30-Mar	6-Apr			
9.1	Executive Summary Development	3					
9.2	Report Sections Integration	4					
9.3	Schedule Assessment & EVM Analysis	3					
9.4	Lessons Learned Documentation	2					
9.5	Next Steps Development	2					
9.6	Draft Final Report Cleanup & Submittal	2					
10	**Final Report & Oral Presentation**	11	6-Apr	25-Apr			
10.1	Final Report Revisions	2					
10.2	Presentation Development	3					
10.3	Presentation Materials Preparation	2					
10.4	Final Report & Presentation Submittal	2					
10.5	Oral Presentation Delivery	2					
		292					

Figure 18: Work Breakdown Structure Update

EVM

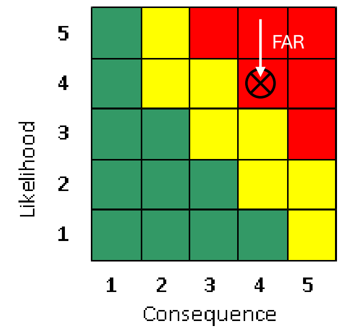
Table 3: EVM Update

					EVM Analysis					
Week	**End of Week Date**	**BCWS**	**BCWP**	**ACWP**	**Cost Variance**	**Schedule Variance**	**SPI**	**CPI**	**TCPI (BAC)**	**EAC**
1	1/26/2025	8	50	5	45	42	6.25	10.00	0.84	29.2
2	2/2/2025	26	50	30	20	24	1.92	1.67	0.92	175.2
3	2/9/2025	44	83.75	55	28.75	39.75	1.90	1.52	0.88	191.8
4	2/16/2025	62	83.75	65	18.75	21.75	1.35	1.29	0.92	226.6
5	2/23/2025	80	83.75	80	3.75	3.75	1.05	1.05	0.98	278.9

Appendix B – Risk Updates

Risk 1 – Navigation System Failure

Risk Title	Navigation System Failure	
Description:	IF the GPS integration fails to meet accuracy requirements THEN autonomous positioning during refueling will be compromised, leading to unsafe operating conditions.	
Initial Assessment:	Likelihood:	5
	Consequences:	4
	Description of Consequences if realized	- Autonomous positioning failures would make it difficult to establish rough positioning of the tanker aircraft. Putting the mission, crew members, and others aboard at risk

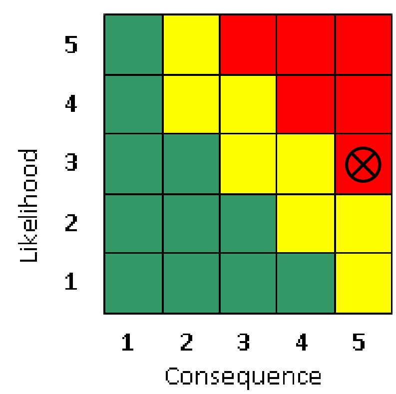


Risk 1: Navigation System Accuracy

Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			
1	Proposal	--	5	4	Initial assessment
2	RAR	Impose requirements for implementation of backup navigation system. See: ANGARS-22, Inertial Navigation.	5	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
3	FAR	Defined F6: Operate in Adverse Conditions behavior	4	4	Likelihood decreases due to implementation of functionality to switch sensors during conditions assessed as adversarial. Further definition of F6.4 should be done to burn down likelihood further.
--		--	--	--	--

Risk 2 – Communications System Failure

Risk Title	Communication System Failure	
Description:	IF secure data link is lost during operation THEN operational delays or mission failure may occur, potentially requiring manual intervention.	
Initial Assessment:	Likelihood:	3
	Consequences:	5
	Description of Consequences if realized	- Comms going down would result in mission failures. Potential critical situational awareness updates may not be conveyed between ANGARS to ground or receiving aircraft, or vise-versa.



Mitigation Plan			L	C	Impact Description & Rationale
ID	Document	Mitigation Action			

1	Proposal	--	3	5	Initial assessment
2	RAR	Impose requirements defining design-level communication up-time constraints. See: ANGARS-33, Communication Uptime.	3	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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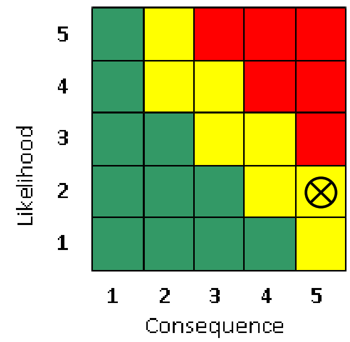
Risk 3 – Collision Avoidance System Failure

Risk Title	**Collision Avoidance System Failure**	
Description:	IF autonomous collision avoidance fails to activate when needed THEN there is an increased risk of mid-air collisions.	
Initial Assessment:	Likelihood:	2
	Consequences:	5
	Description of Consequences if realized	- Autonomous collision avoidance system failures could result in catastrophic consequences such as bird strikes and fuel boom strikes.



Mitigation Plan			**L**	**C**	**Impact Description & Rationale**
ID	**Document**	**Mitigation Action**			
1	Proposal	--	2	5	Initial assessment
2	RAR	Impose requirements defining design-level collision expectations. See: ANGARS-53, Collision Expectancy	2	5	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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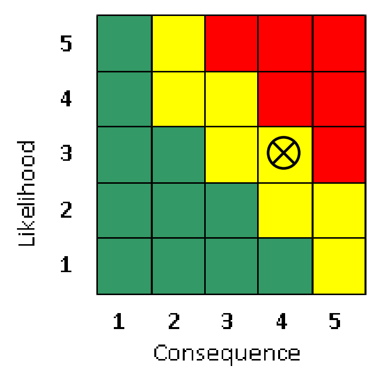
Risk 4 – Integration Challenges

Risk Title	**Collision Avoidance System Failure**	
Description:	IF there are delays in integration with KC-46/NGAS platforms THEN schedule overruns and increased costs will impact project delivery	
Initial Assessment:	Likelihood:	3
	Consequences:	4
	Description of Consequences if realized	- Integration issues could result in significant schedule and cost risks, including overrun/budget. - Potential implications to existing ANGARS architecture or modifications to legacy hardware.



Mitigation Plan	**L**	**C**	**Impact Description & Rationale**

ID	Document	Mitigation Action			
1	Proposal	--	3	4	Initial assessment
2	RAR	Impose requirements defining design-level integration expectations. See: ANGARS-125, KC-46 Integration	3	4	L/C does not change as implementation needs to be designed and added to system architecture to assess L/C impacts.
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Appendix C – References

https://iwalker.people.clemson.edu/RAFT_IROS15_proceedings.pdf

https://www.visual-paradigm.com/VPGallery/diagrams/Activity.html

https://circle.visual-paradigm.com/docs/modaf-tool/operational-viewpoint/ov-5-operational-activity-model/

EN.645.632.82.FA24 Applied Analytics for Model Based Systems Engineering

- Reference model (profiles)
- Prior model (custom profiles/setup)
- Lectures (modeling guidance)

https://www.paloaltonetworks.com/cyberpedia/what-is-a-zero-trust-architecture

https://en.wikipedia.org/wiki/Spread_spectrum